Project: Worked examples
Title: Duct sizing
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1. Scope and method

Calculates duct sizes by the equal-friction or the velocity method, rounded up to standard sizes. It reports the velocity, friction rate and losses of each section and the pressure drop along the index run.

WHAT IT CALCULATES	RELATION
Round duct diameter from the design target	D by Colebrook with the air properties at T (equal friction), or D = √(4Q/πv) (velocity), rounded up to the next standard size
Rectangular duct of equal friction and capacity	D_e = 1.30 (ab)^0.625/(a + b)^0.25
Flat-oval duct of equal friction and capacity	D_e = 1.55 A^0.625/P^0.25, with A = $\pi b^2/4$ + b(a − b) and P = πb + 2(a − b)
Velocity method, rectangular or flat-oval duct with side a fixed	b from the free area A = Q/v (rectangle b = A/a; flat oval the smaller root of (1 − π/4)b^2 − ab + A = 0), rounded up; D_e is kept for the friction only
Friction loss per metre, at the velocity v_e of the equivalent round duct	$\Delta p/L = f \rho v_e^2/(2D_e)$
Fitting losses, at the actual velocity	$\Delta p = \Sigma C \rho v^2/2$
Index-run totals, summed over every section of the run	$\Delta p_{run} = \Sigma(\Delta p/L \cdot L + \Sigma C \rho v^2/2)$

Basis: Duct design (friction chart basis, equivalent diameters, loss coefficients)[1](Ch. 21); Flow of fluids in pipes and ducts[2](§4); Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions[3]; Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions[4].

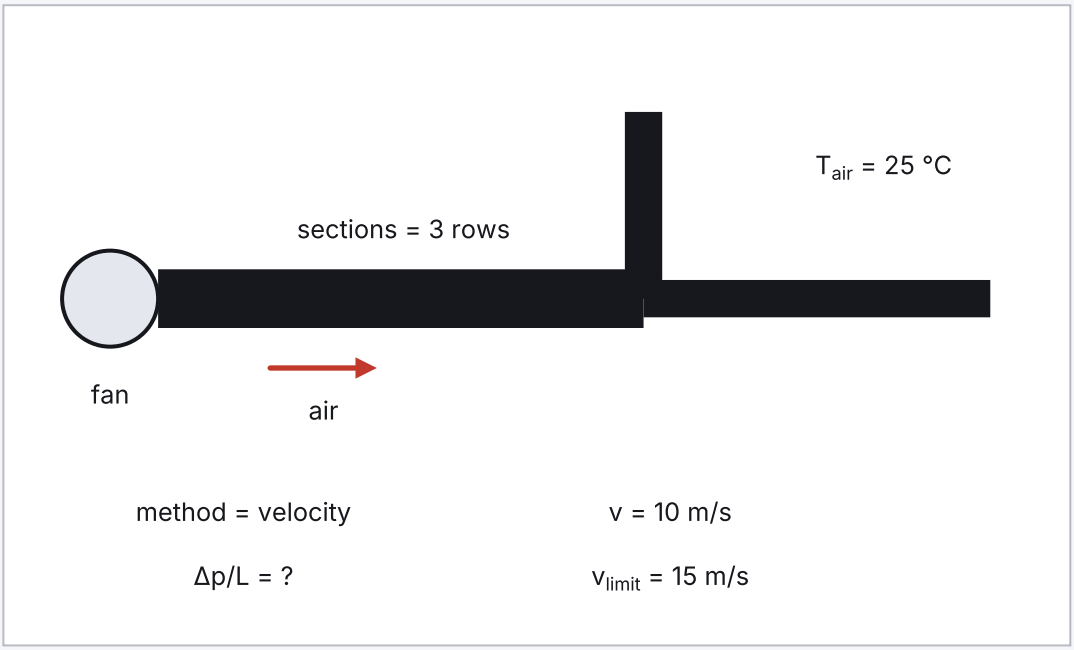


Figure 1 — A duct run from the fan through its sections as listed (a main and two branches drawn), each sized by the chosen method. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Sizing method	velocity		assumption: Velocity method for an extract system.	any
sections	Duct sections	3 row(s)		drawing: DRG-M-310 rev A	any
$round_{to}$	Round sizes up to a multiple of	50	mm	spec: BS EN 1506 preferred sizes	1 … 100 mm
eps_{rough}	Duct roughness	0.15	mm	assumption: Spiral-wound galvanised duct with joints.	0.03 … 3 mm
T_{air}	Air temperature	25	°C	assumption: Workshop air 25 °C.	-20 … 60 °C
p_{atm}	Barometric pressure	100.5	kPa	assumption: Site 70 m AOD, ≈ 100.5 kPa.	80 … 105 kPa
dp_{target}	Target friction rate (equal friction)	1	Pa/m	default	0.3 … 5 Pa/m
v_{target}	Target velocity (velocity method)	10	m/s	spec: Dust-carrying extract: 10 m/s minimum transport velocity, SPEC-M-09	2 … 20 m/s
v_{limit}	Velocity limit (warning and check)	15	m/s	spec: SPEC-M-09 noise limit	2 … 30 m/s

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dp_{limit}	Total pressure-drop limit (check)	600	Pa	spec: Extract fan external static 600 Pa	50 … 2000 Pa

TABLE ROW	VALUES
sections[1]	name = hood branch; Q = 0.35 m³/s; L = 6 m; shape = round; a = —; D = —; C = 1.5; dp_eq = 120 Pa; index = yes
sections[2]	name = main; Q = 1.05 m³/s; L = 20 m; shape = round; a = —; D = —; C = 0.6; dp_eq = 0 Pa; index = yes
sections[3]	name = stack; Q = 1.05 m³/s; L = 8 m; shape = round; a = —; D = —; C = 1; dp_eq = 30 Pa; index = yes

3. Assumptions

A1. Air is dry at the stated temperature and pressure. (Ideal gas, ρ = p/(287.055 T).)
A2. The viscosity of air follows Sutherland's law. (Standard air properties.)
A3. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A4. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A5. method (Sizing method) is taken as velocity. (Velocity method for an extract system.)
A6. eps_{rough} (Duct roughness) is taken as 0.15 mm. (Spiral-wound galvanised duct with joints.)
A7. T_{air} (Air temperature) is taken as 25 °C. (Workshop air 25 °C.)
A8. p_{atm} (Barometric pressure) is taken as 100.5 kPa. (Site 70 m AOD, ≈ 100.5 kPa.)

4. Working

[10] ρ = p / (287.055 T)
= 100500 Pa / (287.055 × 298.15 K)
= 1.1743 kg/m³
Air density

[11] μ = Sutherland's law
= T = 298.15 K
= 18.371 uPa·s
Air dynamic viscosity

[12] D_{s1} = hood branch: diameter
= target 211.1 mm rounded up to 50 mm
= 250 mm
Diameter, hood branch

[13] v_{s1} = hood branch: Q / A
= 0.35 / 0.049087
= 7.1301 m/s
Velocity, hood branch

[14] Re_{s1} = hood branch: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.1743 × 7.1301 × 0.25 / 0.000018371
= 113940
Reynolds number, hood branch

[15] f_{s1} = hood branch: friction factor
= Colebrook–White
= 0.02041
Friction factor, hood branch

[16] dpL_{s1} = hood branch: f ρ v_e^2 / (2 D_e)
= 0.02041 × 1.1743 × 7.1301² / (2 × 0.25)
= 2.4369 Pa/m

Friction rate, hood branch

[17] $dp_{f,s1}$ = hood branch: (Δp/L) L
= 2.4369 × 6
= 14.622 Pa

Friction loss, hood branch

[18] $dp_{fit,s1}$ = hood branch: ΣC ρ v²/2
= 1.5 × 29.849
= 44.774 Pa

Fitting loss, hood branch

[19] dp_{s1} = hood branch: friction + fittings + equipment
= 14.622 + 44.774 + 120
= 179.4 Pa

Section loss, hood branch (index run)

[20] D_{s2} = main: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm

Diameter, main

[21] v_{s2} = main: Q / A
= 1.05 / 0.12566
= 8.3556 m/s

Velocity, main

[22] Re_{s2} = main: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630

Reynolds number, main

[23] f_{s2} = main: friction factor
= Colebrook–White
= 0.018028

Friction factor, main

[24] dpL_{s2} = main: f ρ v_e² / (2 D_e)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m

Friction rate, main

[25] $dp_{f,s2}$ = main: (Δp/L) L
= 1.8475 × 20
= 36.95 Pa

Friction loss, main

[26] $dp_{fit,s2}$ = main: ΣC ρ v²/2
= 0.6 × 40.992
= 24.595 Pa

Fitting loss, main

[27] dp_{s2} = main: friction + fittings + equipment
= 36.95 + 24.595 + 0
= 61.545 Pa

Section loss, main (index run)

[28] D_{s3} = stack: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm

Diameter, stack

[29] v_{s3} = stack: Q / A
= 1.05 / 0.12566
= 8.3556 m/s

Velocity, stack

[30] Re_{s3} = stack: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630
Reynolds number, stack
[31] f_{s3} = stack: friction factor
= Colebrook–White
= 0.018028
Friction factor, stack
[32] dpL_{s3} = stack: f ρ v_e^2 / (2 D_e)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, stack
[33] $dp_{f,s3}$ = stack: (Δp/L) L
= 1.8475 × 8
= 14.78 Pa
Friction loss, stack
[34] $dp_{fit,s3}$ = stack: ΣC ρ v^2/2
= 1 × 40.992
= 40.992 Pa
Fitting loss, stack
[35] dp_{s3} = stack: friction + fittings + equipment
= 14.78 + 40.992 + 30
= 85.772 Pa
Section loss, stack (index run)
[36] dp_{index} = Σ section losses on the index run
= 3 section(s)
= 326.71 Pa
Index-run pressure drop
[37] dp_{all} = Σ section losses, all sections
= 3 section(s)
= 326.71 Pa
Sum over every section (not a path)
[38] v_{peak} = highest section velocity
=
= 8.3556 m/s
Highest velocity
[39] u_{dp} = dp_{index} / dp_{limit}
= 326.71 / 600
= 0.54452
Pressure-drop utilisation
[40] u_v = v_{peak} / v_{limit}
= 8.3556 / 15
= 0.55704
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
dp_{index}	326.7 Pa	Index-run Δp
v_{peak}	8.356 m/s	Highest velocity (governing)
D_{s1}	250 mm	D, hood branch
v_{s1}	7.130 m/s	v, hood branch
Re_{s1}	113940	Re, hood branch

SYMBOL	VALUE	DESCRIPTION
f_{s1}	0.02041	f, hood branch
dpL_{s1}	2.437 Pa/m	Δp/L, hood branch
$dp_{f,s1}$	14.62 Pa	Friction loss, hood branch
$dp_{fit,s1}$	44.77 Pa	Fitting loss, hood branch
dp_{s1}	179.4 Pa	Section loss, hood branch
D_{s2}	400 mm	D, main
v_{s2}	8.356 m/s	v, main
Re_{s2}	213630	Re, main
f_{s2}	0.018028	f, main
dpL_{s2}	1.848 Pa/m	Δp/L, main
$dp_{f,s2}$	36.95 Pa	Friction loss, main
$dp_{fit,s2}$	24.59 Pa	Fitting loss, main
dp_{s2}	61.55 Pa	Section loss, main
D_{s3}	400 mm	D, stack
v_{s3}	8.356 m/s	v, stack
Re_{s3}	213630	Re, stack
f_{s3}	0.018028	f, stack
dpL_{s3}	1.848 Pa/m	Δp/L, stack
$dp_{f,s3}$	14.78 Pa	Friction loss, stack
$dp_{fit,s3}$	40.99 Pa	Fitting loss, stack
dp_{s3}	85.77 Pa	Section loss, stack
dp_{all}	326.7 Pa	Σ all sections

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Highest velocity against limit	8.356 m/s	15 m/s	0.56	
PASS	Index-run pressure drop against limit	326.7 Pa	600 Pa	0.54	

7. Applicability

None.

Defaults used (inputs left empty):
dp_{target}: not entered, so the default of 1 Pa/m is used.

8. Verification

```
Template bs.duct-sizing@1.2.1 Content hash a0afb84d934b74eb Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
3. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Rectangular Cross-Section—Dimensions. BS EN 1505. BSI.
4. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Circular Cross-Section—Dimensions. BS EN 1506. BSI.